

AI-Powered Phishing Detection and Simulation Platform

Live Industry Project – Technical Proposal and System Documentation

Project Overview

The AI-Powered Phishing Detection and Simulation Platform is a cybersecurity system designed to help organizations detect phishing attacks and improve employee awareness through controlled phishing simulations. Phishing remains one of the most common attack methods used by cybercriminals to gain unauthorized access to sensitive information. Employees frequently receive emails that appear legitimate but contain malicious links or credential-harvesting pages.

This project proposes a centralized platform that combines artificial intelligence, secure cloud infrastructure, and a modern web dashboard to detect phishing attempts and simulate phishing attacks within an organization. The platform will analyze email content and URLs using machine learning techniques while also allowing administrators to run phishing simulation campaigns to evaluate employee security awareness.

The platform provides both defensive security capabilities and cybersecurity awareness training. It helps organizations identify phishing threats and simultaneously train employees to recognize malicious emails, thereby reducing the overall risk of cyber attacks.

Problem Statement

Phishing attacks continue to be one of the leading causes of security breaches in organizations. Attackers often impersonate trusted services such as banks, internal IT departments, or cloud services to trick users into revealing confidential information. Traditional spam filters are often unable to detect sophisticated phishing campaigns that mimic legitimate communication.

Organizations also face challenges in measuring employee vulnerability to phishing attacks. Without proper training and testing mechanisms, employees remain susceptible to social engineering techniques. As a result, organizations require a system that can both detect phishing attempts and provide controlled phishing simulations to improve employee awareness.

The proposed platform addresses these challenges by integrating artificial intelligence-based detection with phishing simulation tools and security analytics.

Project Objectives

The main objective of this project is to develop a secure platform capable of detecting phishing attacks and providing phishing awareness training for organizations.

The system aims to achieve the following objectives:

- Detect phishing emails and malicious URLs using machine learning models
- Provide a phishing simulation platform for controlled cybersecurity training
- Monitor employee interaction with phishing simulation emails
- Generate security analytics and reports for administrators
- Calculate employee risk scores based on simulation responses

- Improve cybersecurity awareness within organizations

Project Scope

The scope of this project focuses on building a web-based platform that provides phishing detection and simulation capabilities.

The system will analyze email content and URLs using AI models to determine whether they are phishing attempts. It will also provide an administrative interface where security teams can create phishing simulation campaigns and track employee responses.

The platform will store campaign data, detection results, and user activity logs in a cloud database. The system will initially operate as a standalone security training platform and will not directly integrate with enterprise email providers during the first version.

Future versions may include integration with enterprise email systems and advanced threat intelligence sources.

Technology Stack

The platform will be developed using modern web technologies and cloud services to ensure scalability, security, and performance.

Frontend Development

The frontend will be developed using Next.js, a modern React-based framework that supports both server-side and client-side rendering. The dashboard will provide an intuitive interface for administrators to manage phishing campaigns, analyze detection results, and review employee behavior metrics.

Tailwind CSS will be used for styling and responsive design, while chart libraries such as Chart.js or Recharts will be used to display security analytics and campaign performance reports.

Backend Development

The backend logic will be implemented using Next.js API routes running on a Node.js environment. The backend will handle authentication, phishing campaign management, data processing, AI detection requests, and communication with the database.

RESTful APIs will enable communication between the frontend dashboard and backend services.

Artificial Intelligence Layer

The phishing detection engine will be developed using Python with machine learning libraries. The AI layer will analyze email text, suspicious keywords, and URL structures to identify phishing patterns.

Machine learning models such as Logistic Regression or Random Forest will be used for the initial implementation. Natural language processing techniques will be used to extract features from email content and detect suspicious language patterns commonly found in phishing messages.

Cloud Infrastructure and Database

The system will use Firebase services for secure cloud infrastructure.

Firebase Authentication will provide secure login and identity management. Firestore will serve as the primary cloud database to store users, campaign data, detection logs, and analytics results. Firebase Cloud Functions may be used for serverless processing of background tasks. Firebase Hosting will be used to deploy the web platform.

System Architecture

The AI phishing detection platform follows a modular architecture composed of multiple interacting components that support phishing simulation, detection, and user behavior analysis.

The **Admin Dashboard** serves as the frontend interface for security administrators. Through this dashboard, administrators can configure phishing campaigns, monitor system analytics, and review detection results and user behavior reports.

The **Backend API layer** acts as the core application logic of the platform. It handles requests from the frontend dashboard, manages phishing campaigns, communicates with the AI detection engine, and coordinates data storage operations.

The **Email Campaign Engine** is responsible for generating and distributing simulated phishing emails to employees. It integrates with the **Phishing Email Generator**, which uses a Large Language Model (LLM) to produce realistic phishing messages for training and testing purposes.

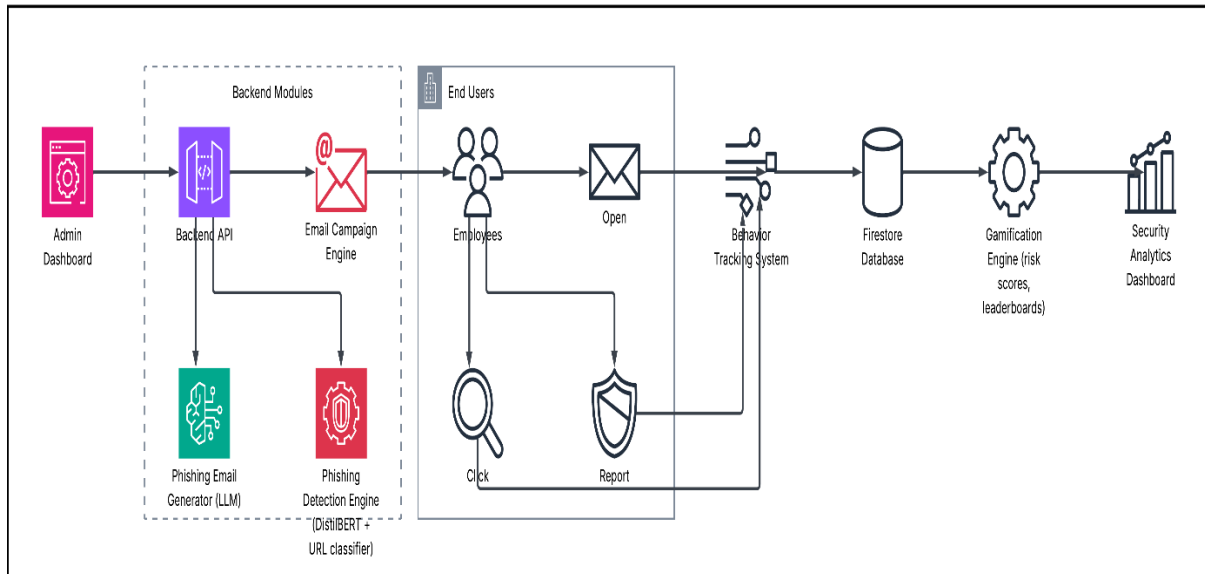
The **AI Detection Engine** analyzes email content and embedded URLs to determine whether messages are malicious or legitimate. This component utilizes machine learning models including **DistilBERT for text analysis** and a **URL classification model** for identifying suspicious links.

The **End User Interaction Layer** represents employee actions such as opening emails, clicking links, or reporting suspicious messages. These events are captured by the Behavior Tracking System and stored for further analysis, which records user responses to phishing simulations.

All platform data, including user records, phishing campaigns, detection results, and behavioral logs, are stored in the **Firestore database**.

Finally, the **Gamification Engine** processes behavioral data to generate risk scores, training feedback, and leaderboard rankings, which are visualized through the **Security Analytics Dashboard** to help organizations assess employee security awareness.

AI Phishing Detection Platform Architecture



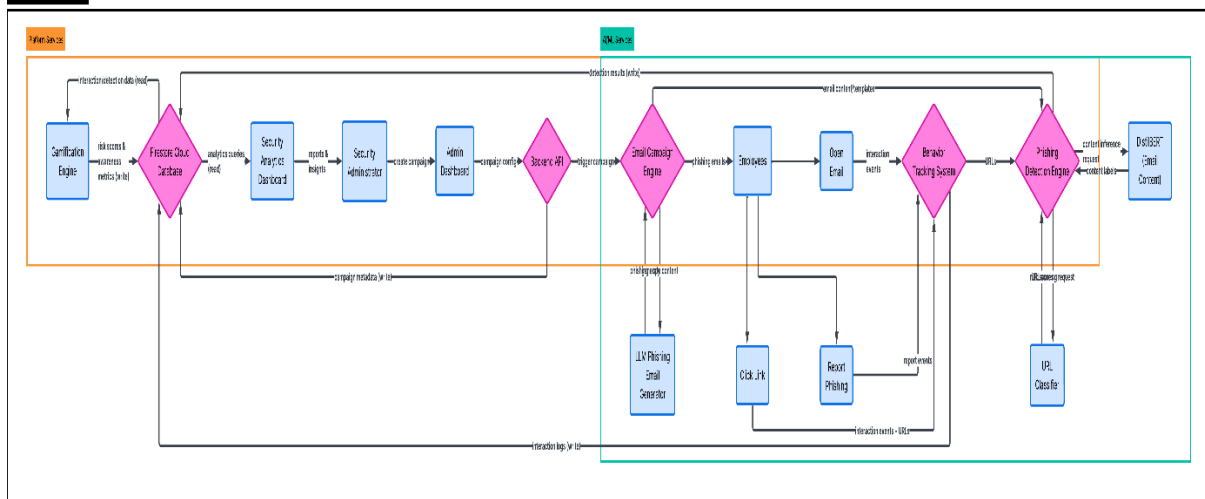
System Workflow

The phishing detection workflow begins when an email or URL is submitted to the platform for analysis. The backend API sends the content to the AI detection service, which processes the text using machine learning models. The AI model returns a prediction indicating whether the message is phishing along with a confidence score. The backend stores the result in the database and displays it in the security dashboard.

For phishing simulation campaigns, administrators create campaigns using predefined templates such as password reset requests or invoice notifications. The system sends simulated phishing emails to selected employees. Each email contains a tracking link that records user interactions.

When employees interact with the email, the system logs their actions, including opening the email, clicking links, submitting credentials, or reporting the phishing attempt. These interactions are analyzed to generate employee risk scores and campaign analytics.

System Workflow Diagram



Main Platform Modules

AI Phishing Detection Module:

This module analyzes emails and URLs to determine whether they are phishing attempts. The AI model processes text patterns, suspicious keywords, domain structures, and URL characteristics to identify malicious messages.

Phishing Simulation Module:

This module allows administrators to create phishing simulation campaigns that mimic real-world phishing scenarios. Email templates can represent common attack types such as password resets, invoice requests, or system alerts.

User Behavior Tracking Module:

This module records employee interactions with simulated phishing emails. Actions such as email opening, link clicking, credential submission, and phishing reporting are logged for analysis.

Security Dashboard:

The dashboard provides administrators with a centralized interface to monitor phishing detection results, campaign performance, and employee risk levels. Security teams can analyze metrics such as click rates, reporting rates, and overall phishing vulnerability.

Gamification Module

This module processes behavioural data collected during phishing simulations and calculates employee risk scores. It generates leaderboards, training feedback, and security awareness levels to encourage employees to recognize phishing attempts and improve cybersecurity practices.

Database Architecture

The platform will store all data using Firebase Firestore with structured collections.

The Users collection stores employee and administrator account details including name, email, role, department, and risk score.

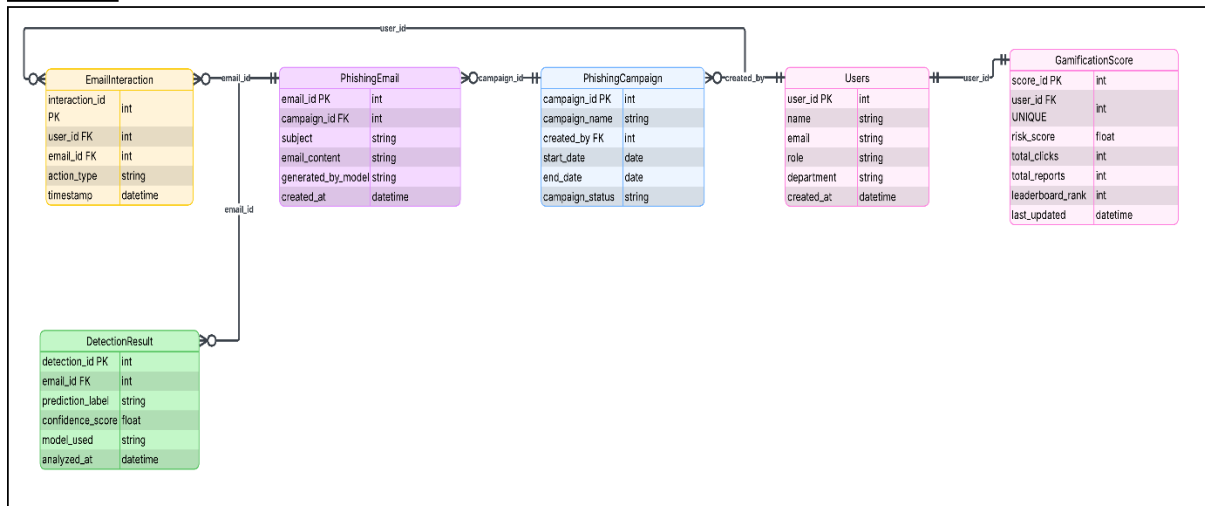
The Campaigns collection stores phishing campaign configurations including campaign name, target users, template type, and campaign status.

The Emails collection stores records of emails sent during phishing simulations including recipient information and interaction results.

The Detection Logs collection stores results generated by the AI phishing detection engine including prediction results and confidence scores.

The Click Tracking collection records employee interactions with phishing links and includes details such as timestamp, device information, and campaign identifiers.

The Risk Score collection stores calculated security risk levels for employees based on their behavior during phishing campaigns.



User Roles

The platform will support different user roles with controlled access.

Administrators are responsible for managing phishing campaigns, monitoring analytics, and controlling system settings.

Security Analysts review phishing detection results, analyze employee behavior, and generate security reports.

Employees receive simulated phishing emails and interact with them as part of awareness training.

Security Implementation

Security is a key component of the platform.

Firebase Authentication will provide secure login and identity management for all users.

Role-based access control will restrict administrative functions to authorized users.

Firestore security rules will limit database access based on authenticated roles.

All communication between system components will be encrypted using HTTPS.

Backend APIs will validate and sanitize inputs to prevent injection attacks and malicious data submissions.

Development Plan

The project will be implemented in structured phases.

The first phase focuses on system design and architecture planning. During this phase, the team will finalize the technology stack, configure Firebase services, and establish the project repository.

The second phase involves developing the core system components including the backend APIs, AI detection model, and frontend dashboard.

The third phase focuses on integrating the AI detection service with the backend and implementing phishing simulation features.

The final phase involves system testing, security validation, performance optimization, and deployment.

Expected Outcome

The completed platform will provide organizations with a practical cybersecurity tool capable of detecting phishing threats and improving employee awareness through phishing simulations.

Security teams will be able to identify high-risk employees, analyze phishing campaign performance, and strengthen their organization's security posture.

The system will combine artificial intelligence and security awareness training to create a proactive defense mechanism against phishing attacks.

Future Enhancements

Future improvements may include integration with enterprise email systems such as Gmail and Microsoft Outlook, development of a browser extension for real-time phishing detection, deployment of advanced NLP models such as transformer-based architectures, and integration with security monitoring platforms such as SIEM systems.

These enhancements will further strengthen the platform's ability to detect and prevent phishing attacks within organizations.